DO YOU KNOW?

Puzzles are instrumental for the origin of Algebra. **Leelavathi** is the first puzzle book in India written by Baskaracharya (Baskara II) who lived in Maharashtra during 12th century. The modified version of the interesting puzzle is for you.

"The beads from a pearl necklace fell down. One third of the pearls fell on the ground. One fifth of the pearls rolled under cot. Two persons started collecting the pearls. One person was able to collect one sixth and the other person collected one tenth of the pearls. If only 6 pearls are left in the necklace, find the total number of pearls in the necklace?"

Let the total number of pearls be x.

Using the given details, we can create an equation.

$$x-\left(\frac{x}{3}+\frac{x}{5}+\frac{x}{6}+\frac{x}{10}\right) = 6$$

$$x-x\left(\frac{10}{30}+\frac{6}{30}+\frac{5}{30}+\frac{3}{30}\right) = 6$$

LCM taken for 5,6,3,10 is 30.

So,
$$x\left(1-\frac{24}{30}\right) = 6$$

$$x\left(\frac{30-24}{30}\right) = 6$$

$$x\left(\frac{6}{30}\right) = 6$$

$$x\left(\frac{1}{5}\right) = 6$$

$$x = 6 \times 5 = 30 \text{ pearls}$$



Summary

- A symbol having a fixed numerical value is called a constant, and a symbol which takes various numerical values is called a variable.
- Variables and constants are combined by the operations addition and subtraction to construct an algebraic expression.
- Any mathematical formula, rule or pattern can be generalized using an algebraic expression.
- The parts of an algebraic expression which are combined by the signs '+' or '−' are called the terms of the expressions.
- In a term of an algebraic expression, each factor or group of factors is called the co-efficient of the remaining factors. The number in the variable term is called the numerical co-efficient.
- The terms having the same algebraic factors are called like terms. Terms with different algebraic factors are called unlike terms.
- We can add or subtract like terms only. Unlike terms cannot be added or subtracted.
- An equation can be looked as algebraic expression equated either to a constant or to any other algebraic expression.
- To solve an equation, we can add or subtract or multiply or divide the same number on both sides of the equation.

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